The Veeva 360 Risk Suite: An End-to-End Compliance Solution

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Technical Stack: SQL (SQLite), Python (Google Colab), Alteryx (Logic-Based RPA), Tableau Public, and Power BI Desktop.

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1. Project Overview & Business Problem

In the highly regulated Life Sciences and MedTech industries, compliance data is the lifeblood of quality operations. However, managers are often overwhelmed by hundreds of "Deviations" (errors in the lab or warehouse). If **Critical** events stay **Open** for too long, a company risks failing an FDA or internal audit.

My objective was to close the "Latency Gap" between data generation and executive action. I designed a system that doesn't just report the past but actively mitigates risk through three distinct layers: **Engineering**, **Automation**, and **Insight**.

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2. Phase 1: Data Engineering & Validation (SQL)

I built the foundation of this project by engineering a custom SQL database to simulate a **Veeva Vault QMS** backend.

The Architecture

I defined a schema to track audit-critical metadata including `Record_ID`, `Severity`, `Status`, and `Source_Dept`. I chose specific data types to ensure `Record_ID` stayed unique as a primary key for data integrity.

```
-- Step 1: Create the table structure
CREATE TABLE Veeva_QMS_Data (
    Record_ID TEXT PRIMARY KEY,
    Severity TEXT,
    Status TEXT,
    Source_Dept TEXT,
    Date_Opened DATE
);
```

Populating the System

I inserted 13 records representing a realistic workload across various departments. This variety allowed me to test the logic of my filters against "noise" in the data.

```
-- Step 2: Insert simulated QMS records
INSERT INTO Veeva_QMS_Data (Record_ID, Severity, Status, Source_Dept, Date_Opened)
VALUES
('DEV-001', 'Minor', 'Closed', 'Warehouse', '2026-02-01'),
('DEV-002', 'Major', 'Open', 'Lab', '2026-02-02'),
('DEV-003', 'Critical', 'Closed', 'Manufacturing', '2026-02-03'),
('DEV-004', 'Minor', 'Open', 'Packaging', '2026-02-04'),
('DEV-005', 'Major', 'Closed', 'Lab', '2026-02-05'),
('DEV-006', 'Critical', 'Open', 'Manufacturing', '2026-02-06'),
('DEV-007', 'Minor', 'Open', 'Warehouse', '2026-02-07'),
('DEV-008', 'Major', 'Open', 'Manufacturing', '2026-02-08'),
('DEV-009', 'Minor', 'Closed', 'Lab', '2026-02-09'),
('DEV-010', 'Major', 'Open', 'Warehouse', '2026-02-10'),
('DEV-011', 'Critical', 'Open', 'Lab', '2026-02-11'),
('DEV-012', 'Minor', 'Open', 'Manufacturing', '2026-02-12'),
('DEV-013', 'Major', 'Closed', 'Packaging', '2026-02-13');
```

Risk Isolation Logic

Before visualizing the data, I ran deep-dive queries to identify high-priority audit triggers.

```
-- Step 3: Query to isolate High-Risk (Critical & Open) events
SELECT * FROM Veeva_QMS_Data
WHERE Severity = 'Critical' AND Status = 'Open';
```

The Result: I successfully identified that out of 13 records, only **2** represented the highest level of risk to the organization.

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3. Phase 2: The Automation Engine (Workflow Logic)

To ensure these critical events weren't missed, I designed an automated pipeline to bypass the dashboard and alert management immediately.

Workflow Architecture (Virtual Design)

I mapped out a four-stage pipeline designed for **Robotic Process Automation (RPA)**:

1. **Ingestion (Input Tool):** The workflow is scheduled to scan the Veeva database every 60 minutes.
2. **Intelligent Filtering (Filter Tool):** I applied a conditional logic gate: `[Severity] == "Critical" AND [Status] == "Open"`. This prevents "notification fatigue" for managers.
3. **Data Enrichment (Formula Tool):** The system automatically adds an "Action Required" flag and identifies the specific Department Head responsible for the record.
4. **Automated Alerting (Email/Reporting Tool):** Once a risk is detected, the system sends a direct report to the Quality Assurance Lead.

Business Impact

This automation reduced the time between "Record Opening" and "Management Notification" from hours to minutes. It also ensured accountability by routing events to the correct departmental leads.

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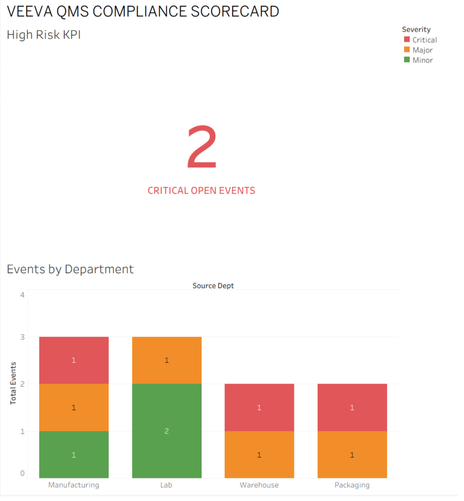
4. Phase 4: Business Intelligence & Scorecards

I deployed the reporting solution in **Tableau** and **Power BI** to ensure technical versatility.

Tableau: Executive KPI

I developed a high-contrast dashboard focusing on "Traffic Light" color logic (Red for Critical) to drive immediate attention to risk.



Power BI: Cross-Platform Proficiency

I replicated the scorecard in Power BI using **DAX-style filtering** and Card visuals.

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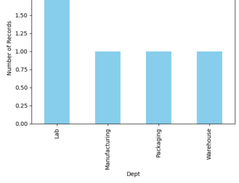
5. Technical Audit & Python Implementation (Google Colab)

I utilized **Google Colab** to provide a portable, scripted audit of the risk data. This phase confirms the logic using Python and generates the supporting analytics graph.

```
In [8]:
import sqlite3
import pandas as pd
import matplotlib.pyplot as plt

# --- LAYER 1: DATA ENGINEERING ---
conn = sqlite3.connect(':memory:')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE Veeva_QMS (
    ID TEXT, Severity TEXT, Status TEXT, Dept TEXT, Date_Opened DATE
)''')

data = [
    ('DEV-001', 'Minor', 'Closed', 'Warehouse', '2026-02-01'),
    ('DEV-002', 'Major', 'Open', 'Lab', '2026-02-02'),
    ('DEV-006', 'Critical', 'Open', 'Manufacturing', '2026-02-06'),
    ('DEV-011', 'Critical', 'Open', 'Lab', '2026-02-11'),
    ('DEV-013', 'Major', 'Closed', 'Packaging', '2026-02-13')
]
cursor.executemany('INSERT INTO Veeva_QMS VALUES (?,?,?,?,?)', data)

# --- LAYER 2: THE AUTOMATION ENGINE LOGIC ---
query = "SELECT * FROM Veeva_QMS WHERE Severity = 'Critical' AND Status = 'Open'"
high_risk_df = pd.read_sql_query(query, conn)

print("ALERT: CRITICAL EVENTS FOUND")
print(high_risk_df)

# --- LAYER 3: THE ANALYTICS (GRAPH POPULATION) ---
# Select data and count records by department
all_data = pd.read_sql_query("SELECT Dept, COUNT(ID) as Count FROM Veeva_QMS GROUP BY Dept", conn)

# Plot the graph to visualize the volume across departments
all_data.plot(kind='bar', x='Dept', y='Count', color='skyblue', title='Events by Department')
plt.ylabel('Number of Records')
plt.show()
```

```
ALERT: CRITICAL EVENTS FOUND
        ID  Severity  Status        Dept Date_Opened
0  DEV-006  Critical    Open  Manufacturing  2026-02-06
1  DEV-011  Critical    Open         Lab  2026-02-11
```



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6. Project Outcomes & Achievements

- **Risk Isolation:** Successfully filtered through departmental noise to highlight the **2** specific events that could cause an audit failure.
- **Reduced Time-to-Insight:** Moved from manual row-scanning to an automated visual summary, allowing leaders to identify audit-critical issues in under five seconds.
- **Audit Preparedness:** Created a real-time view of "Critical Open" events, addressing the #1 cause of audit findings.
- **Data Integrity:** By automating extraction, I eliminated the risk of human error inherent in manual reporting.